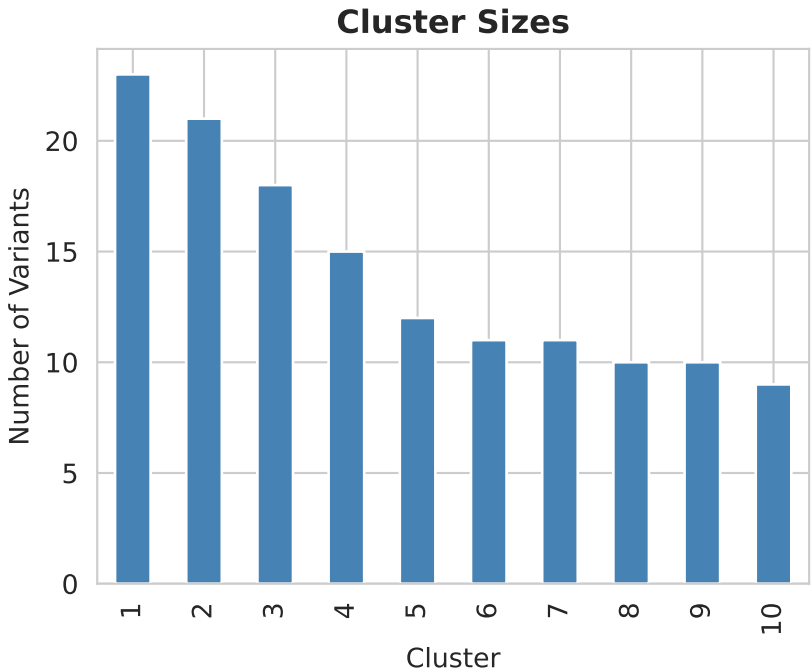
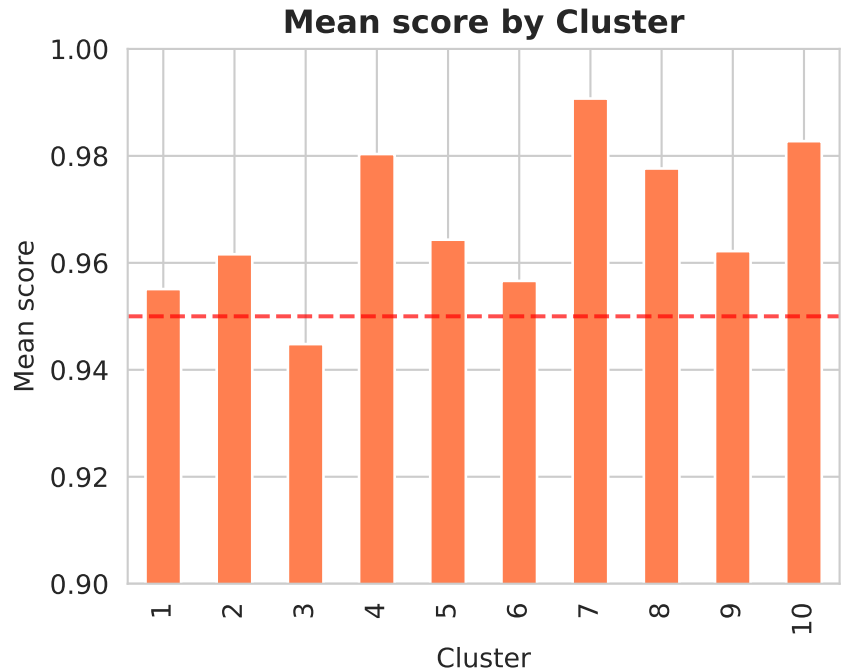
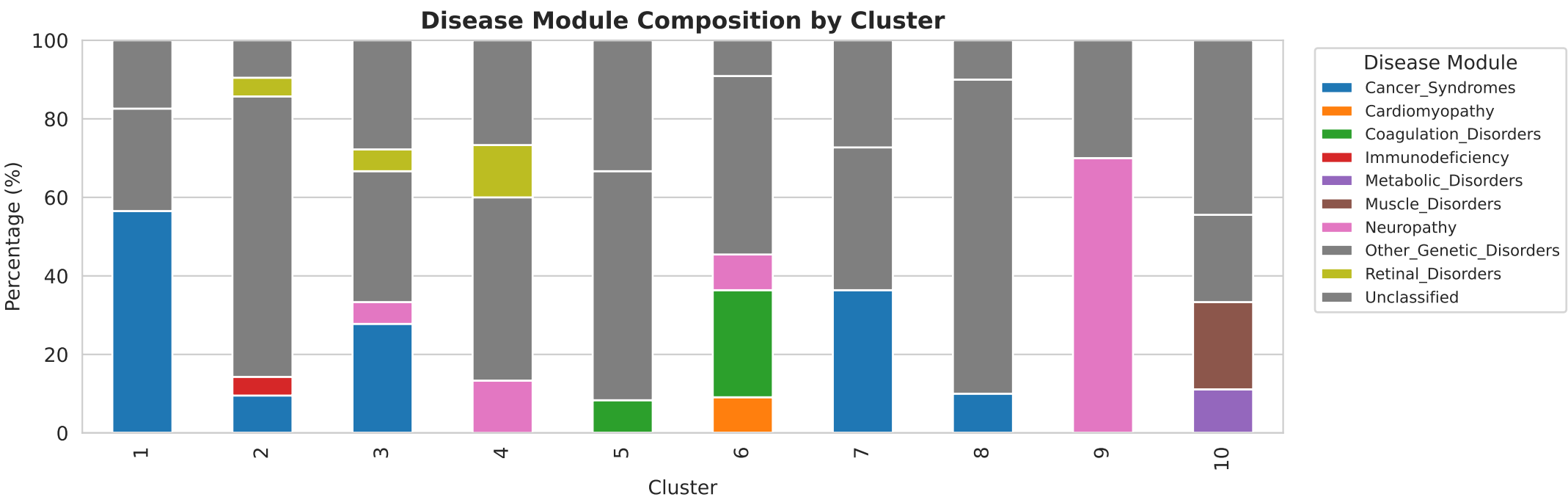
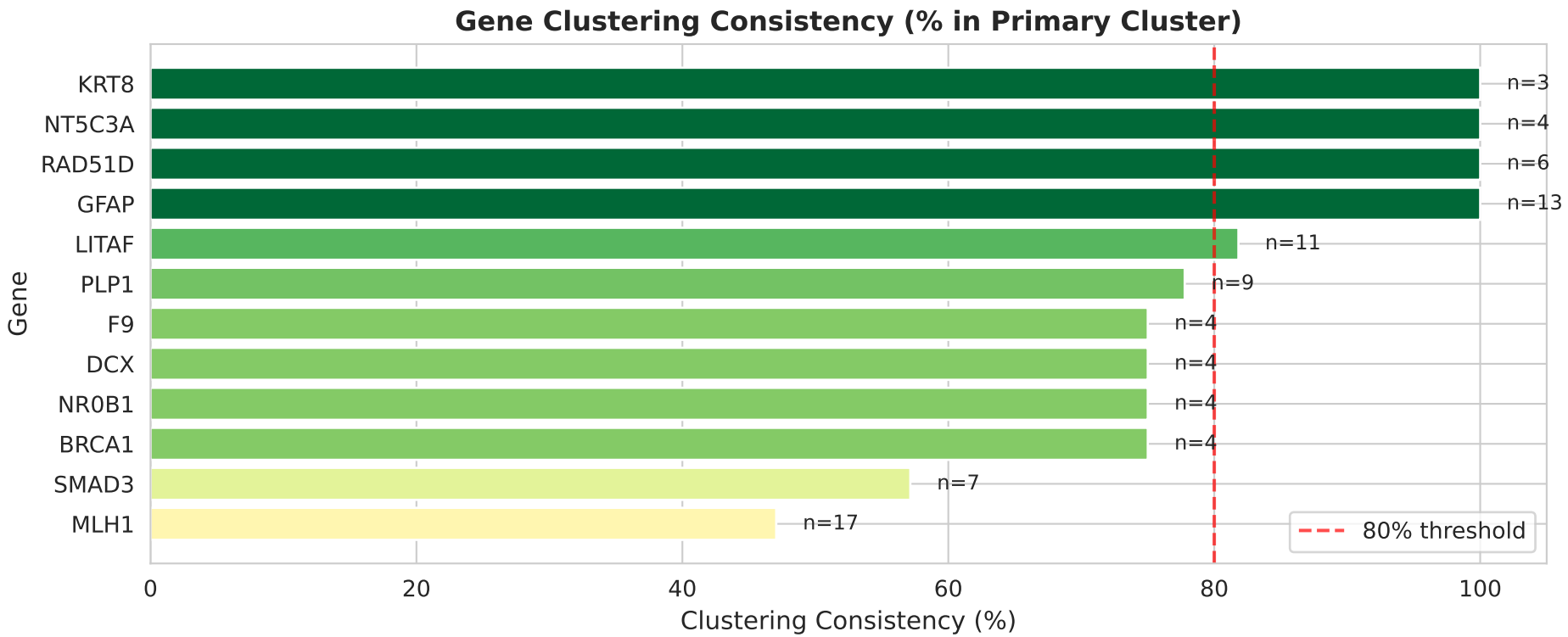
Comprehensive Disease Biology Analysis - Overview

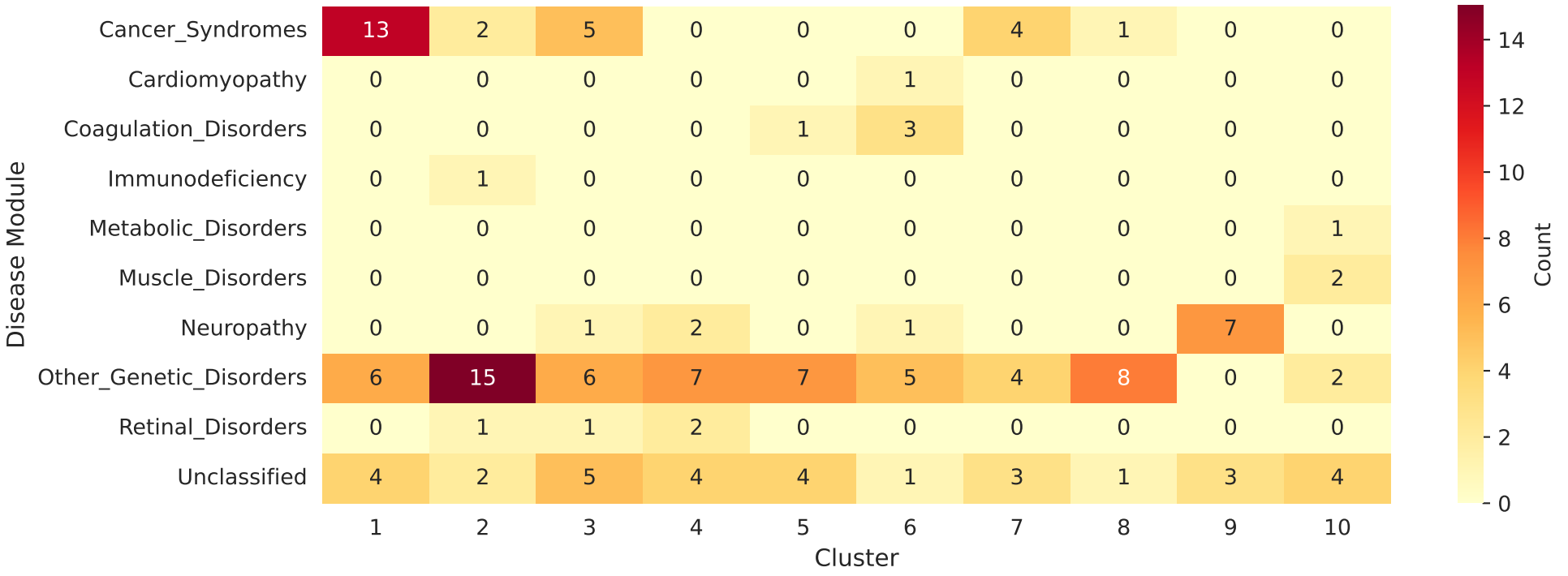






Gene and Disease Module Patterns





Cluster 1: Comprehensive Analysis

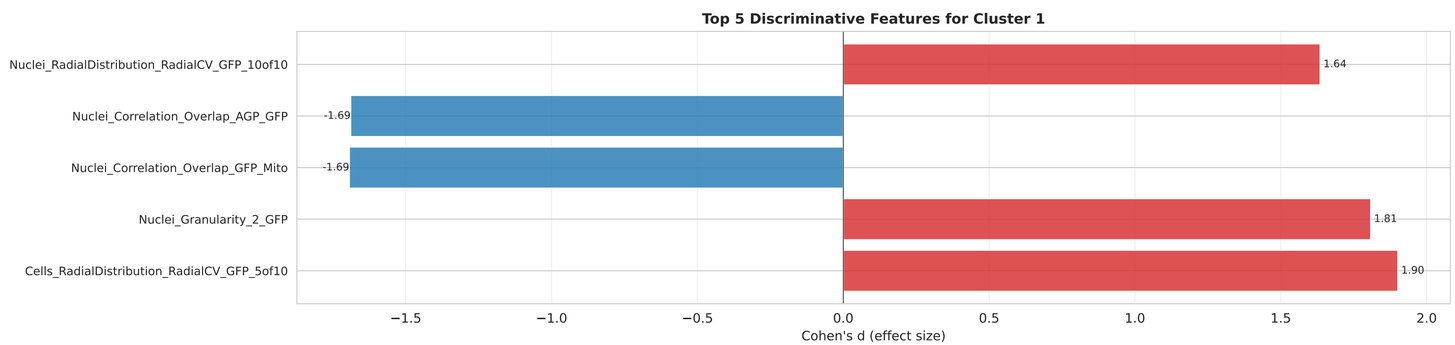
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CLUSTER 1 ANALYSIS
================================================================================================

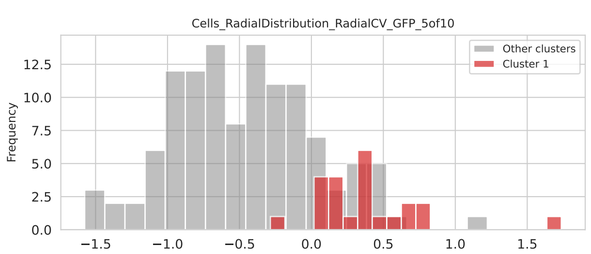
Size: 23 variants
Mean score: 0.955
Hit variants: 17 (73.9%)

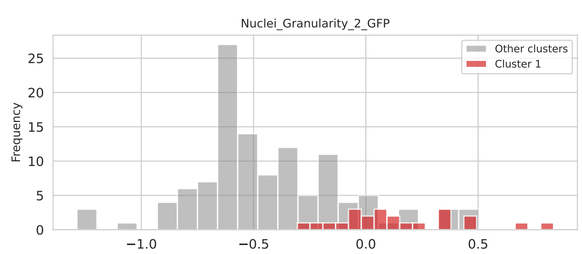
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BFSP2_Arg287Trp         |  BFSP2_Arg339His         |  KRT8_Tyr54His          |  KRT8_Ile63Val
  KRT8_Gly434Ser          |  MLH1_His318Gln          |  MLH1_Ser508Arg         |  MLH1_Lys618Arg
  MLH1_Pro640Ser          |  MLH1_Lys670Arg          |  MLH1_Cys680Arg         |  MLH1_Ala681Thr
  MLH1_Val716Met          |  MSRB3_Cys82Gly          |  RAD51D_Ile17Val        |  RAD51D_Asn138His
  RAD51D_Ala231Ser        |  RAD51D_Phe305Leu        |  RAD51D_Met308Val       |  RAD51D_Thr328Ile
  SMAD3_Val410Phe         |  ZC4H2_Arg18Lys          |  ZC4H2_Leu66His

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic tumor: 14 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, RAD51D_Met308Val, RAD51D_Ile17Val, RAD51D_Thr328Ile, RAD51D_Phe305Leu, RAD51D_Asn138His, RAD51D_Ala231Ser, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Rare genetic immune disease: 8 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Inherited cancer-predisposing syndrome: 8 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Rare genetic developmental defect during embryogenesis: 2 [ZC4H2_Leu66His, ZC4H2_Arg18Lys]
  Rare genetic neurological disorder: 2 [ZC4H2_Leu66His, ZC4H2_Arg18Lys]
  Rare genetic vascular disease: 1 [SMAD3_Val410Phe]
  Rare genetic systemic or rheumatologic disease: 1 [SMAD3_Val410Phe]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (7x) [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Ser508Arg]
  Disease: BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 4 (6x) [RAD51D_Met308Val, RAD51D_Ile17Val, RAD51D_Thr328Ile, RAD51D_Phe305Leu, RAD51D_Asn138His, RAD51D_Ala231Ser]
  Disease: WIEACKER-WOLFF SYNDROME (2x) [ZC4H2_Leu66His, ZC4H2_Arg18Lys]
  Disease: HEPATITIS C VIRUS, SUSCEPTIBILITY TO (2x) [KRT8_Tyr54His, KRT8_Ile63Val]
  inheritance_pattern: autosomal dominant (13x) [BFSP2_Arg287Trp, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, RAD51D_Met308Val, RAD51D_Ile17Val, RAD51D_Thr328Ile, RAD51D_Phe305Leu, RAD51D_Asn138His, RAD51D_Ala231Ser, MLH1_His318Gln, MLH1_Ser508Arg]
  inheritance_pattern: unknown (6x) [KRT8_Gly434Ser, KRT8_Tyr54His, KRT8_Ile63Val, SMAD3_Val410Phe, MLH1_Lys670Arg, BFSP2_Arg339His]
  inheritance_pattern: x-linked recessive (2x) [ZC4H2_Leu66His, ZC4H2_Arg18Lys]
  pLDDT_Category: >90 (16x) [MLH1_Ala681Thr, BFSP2_Arg287Trp, MSRB3_Cys82Gly, ZC4H2_Leu66His, ZC4H2_Arg18Lys, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Lys618Arg, RAD51D_Met308Val, RAD51D_Ile17Val, RAD51D_Phe305Leu, RAD51D_Asn138His, RAD51D_Ala231Ser, SMAD3_Val410Phe, MLH1_Lys670Arg, MLH1_Ser508Arg]
  pLDDT_Category: ⩽50 (4x) [KRT8_Gly434Ser, KRT8_Tyr54His, KRT8_Ile63Val, RAD51D_Thr328Ile]
  pLDDT_Category: 70-90 (2x) [MLH1_Val716Met, MLH1_His318Gln]
  RSA_Category: 10-90 (12x) [BFSP2_Arg287Trp, MSRB3_Cys82Gly, ZC4H2_Leu66His, ZC4H2_Arg18Lys, KRT8_Gly434Ser, MLH1_Lys618Arg, RAD51D_Met308Val, RAD51D_Phe305Leu, RAD51D_Asn138His, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  RSA_Category: <10 (7x) [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, RAD51D_Ile17Val, RAD51D_Ala231Ser, SMAD3_Val410Phe]
  RSA_Category: ⩾90 (3x) [KRT8_Tyr54His, KRT8_Ile63Val, RAD51D_Thr328Ile]
```







Cluster 2: Comprehensive Analysis

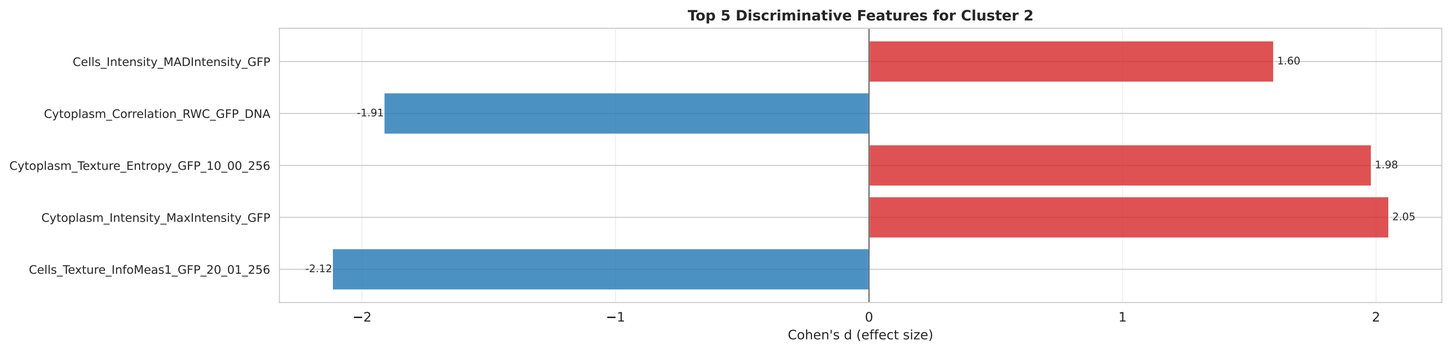
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CLUSTER 2 ANALYSIS
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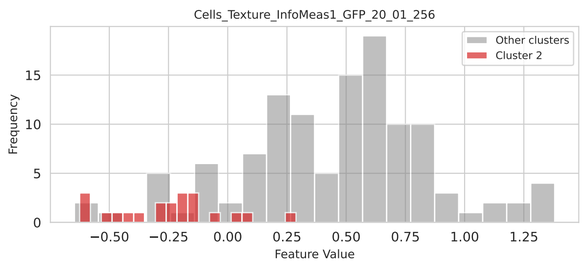
Size: 21 variants
Mean score: 0.962
Hit variants: 20 (95.2%)

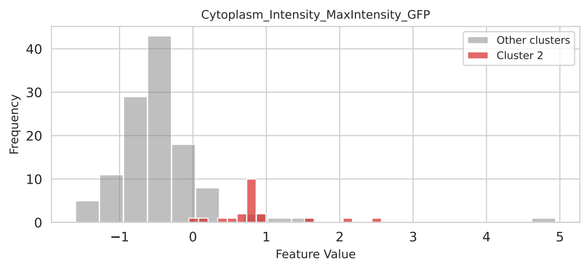
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  APOA1_Trp74Arg          |  APOA1_Val180Glu        |  COMP_Asp319Val         |  GFAP_Asn77Ser
  GFAP_Arg79Cys           |  GFAP_Arg79Ser          |  GFAP_Arg88Cys          |  GFAP_Leu90Pro
  GFAP_Leu235Pro          |  GFAP_Lys236Thr         |  GFAP_Arg239Leu         |  GFAP_Arg239Pro
  GFAP_Lys279Glu          |  GFAP_Leu352Pro         |  GFAP_Tyr366Cys         |  GFAP_Glu371Gly
  IL10_Gly15Arg           |  IMPDH1_Arg154Trp       |  MLH1_Ser44Phe          |  SHOC2_Ser2Gly
  SOS2_Val869Ile

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic eye disease: 15 [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, SHOC2_Ser2Gly, IMPDH1_Arg154Trp, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  Rare genetic neurological disorder: 13 [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  Rare genetic systemic or rheumatologic disease: 2 [APOA1_Val180Glu, APOA1_Trp74Arg]
  Rare genetic renal disease: 2 [APOA1_Val180Glu, APOA1_Trp74Arg]
  Rare genetic endocrine disease: 2 [APOA1_Val180Glu, APOA1_Trp74Arg]
  Rare inborn errors of metabolism: 2 [APOA1_Val180Glu, APOA1_Trp74Arg]
  Rare genetic immune disease: 2 [MLH1_Ser44Phe, IL10_Gly15Arg]
  Rare genetic developmental defect during embryogenesis: 2 [SHOC2_Ser2Gly, COMP_Asp319Val]
  Rare genetic tumor: 1 [MLH1_Ser44Phe]
  Inherited cancer-predisposing syndrome: 1 [MLH1_Ser44Phe]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: ALEXANDER DISEASE (12x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  inheritance_pattern: autosomal dominant (15x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, MLH1_Ser44Phe, GFAP_Leu352Pro, SOS2_Val869Ile, IMPDH1_Arg154Trp, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  inheritance_pattern: unknown (4x) [APOA1_Trp74Arg, IL10_Gly15Arg, COMP_Asp319Val, GFAP_Arg79Ser]
  pLDDT_Category: >90 (13x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, MLH1_Ser44Phe, GFAP_Leu352Pro, SOS2_Val869Ile, COMP_Asp319Val, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Lys279Glu]
  pLDDT_Category: 70-90 (4x) [APOA1_Val180Glu, GFAP_Leu235Pro, GFAP_Glu371Gly, GFAP_Lys236Thr]
  pLDDT_Category: ≤50 (2x) [SHOC2_Ser2Gly, IL10_Gly15Arg]
  RSA_Category: 10-90 (17x) [APOA1_Val180Glu, GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, APOA1_Trp74Arg, GFAP_Leu352Pro, SHOC2_Ser2Gly, COMP_Asp319Val, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  RSA_Category: <10 (2x) [MLH1_Ser44Phe, SOS2_Val869Ile]
```







Cluster 3: Comprehensive Analysis

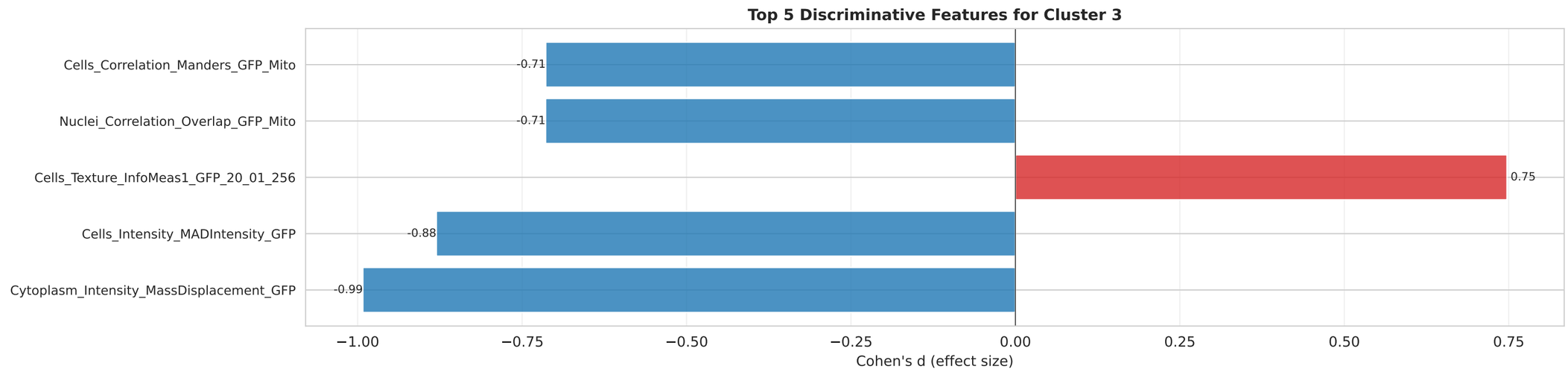
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CLUSTER 3 ANALYSIS
================================================================================================================

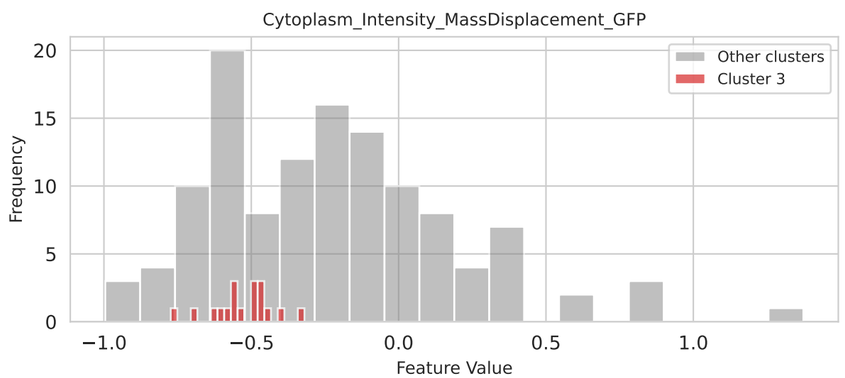
Size: 18 variants
Mean score: 0.945
Hit variants: 10 (55.6%)

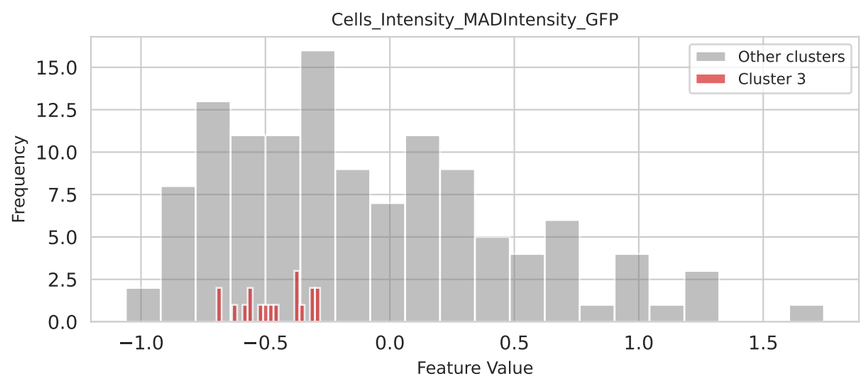
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AGXT_Gly116Arg          |  AGXT_Ile202Asn          |  BRCA1_Met1Ile           |  BRCA1_Met1Thr
  BRCA1_Ala102Gly         |  CDKN1A_Asp149Gly        |  FBP1_Gly164Ser          |  FH_Asn340Lys
  FOXP3_Leu242Pro         |  IMPDH1_Val317Ile        |  LITAF_Ala129Val         |  MLH1_Ser295Thr
  MLH1_Asp485Glu          |  SFTPA2_Gly231Val        |  SMAD3_Trp232Cys         |  SMAD3_Glu239Lys
  SMAD3_Met255Arg         |  SMAD3_Gly272Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic tumor: 7 [BRCA1_Met1Ile, FH_Asn340Lys, BRCA1_Met1Thr, BRCA1_Ala102Gly, MLH1_Ser295Thr, MLH1_Asp485Glu, CDKN1A_Asp149Gly]
  Inherited cancer-predisposing syndrome: 7 [BRCA1_Met1Ile, FH_Asn340Lys, BRCA1_Met1Thr, BRCA1_Ala102Gly, MLH1_Ser295Thr, MLH1_Asp485Glu, CDKN1A_Asp149Gly]
  Rare genetic renal disease: 6 [AGXT_Ile202Asn, BRCA1_Met1Ile, AGXT_Gly116Arg, FH_Asn340Lys, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare inborn errors of metabolism: 4 [AGXT_Ile202Asn, FBP1_Gly164Ser, AGXT_Gly116Arg, FH_Asn340Lys]
  Rare genetic gastroenterological disease: 4 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly, FOXP3_Leu242Pro]
  Rare genetic vascular disease: 4 [SMAD3_Glu239Lys, SMAD3_Gly272Arg, SMAD3_Trp232Cys, SMAD3_Met255Arg]
  Rare genetic systemic or rheumatologic disease: 4 [SMAD3_Glu239Lys, SMAD3_Gly272Arg, SMAD3_Trp232Cys, SMAD3_Met255Arg]
  Rare genetic eye disease: 3 [AGXT_Ile202Asn, AGXT_Gly116Arg, IMPDH1_Val317Ile]
  Rare genetic developmental defect during embryogenesis: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Rare genetic bone disease: 3 [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 1 (3x) [BRCA1_Met1Ile, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  Disease: HYPEROXALURIA, PRIMARY, TYPE I (2x) [AGXT_Ile202Asn, AGXT_Gly116Arg]
  Disease: LOEYS-DIETZ SYNDROME 3 (2x) [SMAD3_Glu239Lys, SMAD3_Gly272Arg]
  inheritance_pattern: autosomal dominant (6x) [SMAD3_Glu239Lys, SMAD3_Gly272Arg, IMPDH1_Val317Ile, MLH1_Ser295Thr, LITAF_Ala129Val, SFTPA2_Gly231Val]
  inheritance_pattern: unknown (5x) [SMAD3_Trp232Cys, SMAD3_Met255Arg, MLH1_Asp485Glu, CDKN1A_Asp149Gly, FOXP3_Leu242Pro]
  inheritance_pattern: multiple (4x) [BRCA1_Met1Ile, FH_Asn340Lys, BRCA1_Met1Thr, BRCA1_Ala102Gly]
  inheritance_pattern: autosomal recessive (3x) [AGXT_Ile202Asn, FBP1_Gly164Ser, AGXT_Gly116Arg]
  pLDDT_Category: >90 (10x) [AGXT_Ile202Asn, FBP1_Gly164Ser, SMAD3_Glu239Lys, AGXT_Gly116Arg, FH_Asn340Lys, SMAD3_Gly272Arg, MLH1_Ser295Thr, SMAD3_Trp232Cys, SMAD3_Met255Arg, SFTPA2_Gly231Val]
  pLDDT_Category: ⩽50 (2x) [BRCA1_Met1Ile, BRCA1_Met1Thr]
  RSA_Category: <10 (8x) [AGXT_Ile202Asn, FBP1_Gly164Ser, SMAD3_Glu239Lys, AGXT_Gly116Arg, MLH1_Ser295Thr, SMAD3_Trp232Cys, SMAD3_Met255Arg, SFTPA2_Gly231Val]
  RSA_Category: ⩾90 (3x) [BRCA1_Met1Ile, BRCA1_Met1Thr, LITAF_Ala129Val]
  RSA_Category: 10-90 (3x) [FH_Asn340Lys, BRCA1_Ala102Gly, SMAD3_Gly272Arg]
```







Cluster 4: Comprehensive Analysis

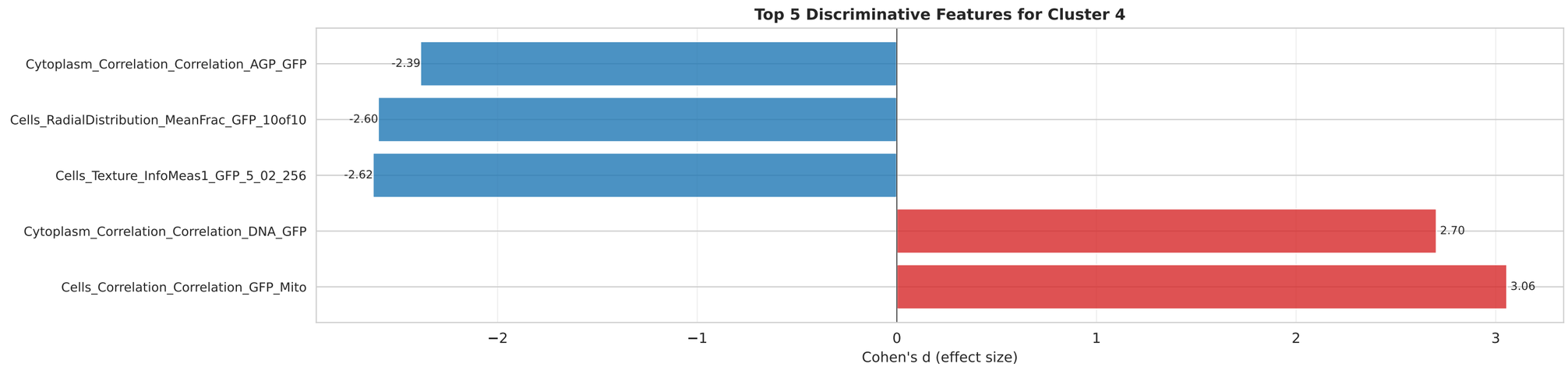
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CLUSTER 4 ANALYSIS
================================================================================================================

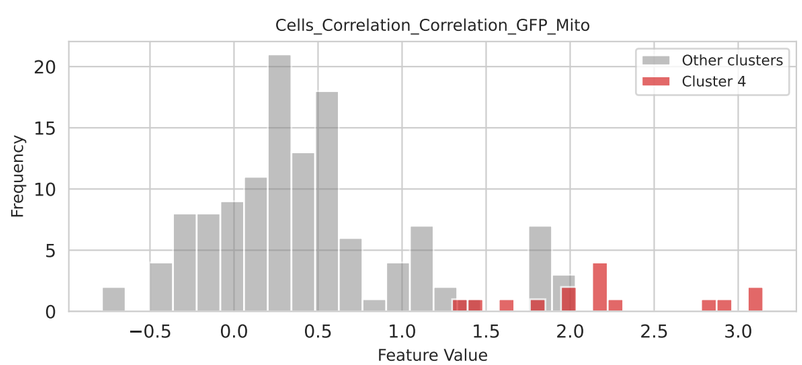
Size: 15 variants
Mean score: 0.980
Hit variants: 13 (86.7%)

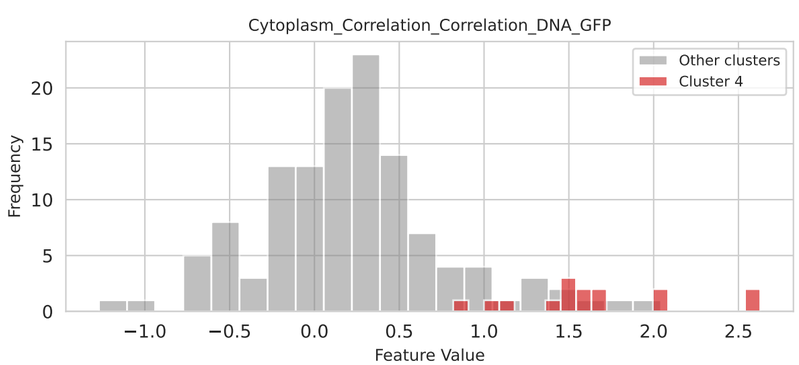
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  PLP1_Thr43Ile           |  PLP1_Thr147Pro         |  PLP1_Ile152Thr         |  PLP1_Asp168His
  PLP1_Gly186Cys          |  PLP1_Leu189Pro         |  PLP1_Cys193Tyr         |  PMP22_Leu16Pro
  PMP22_Leu80Pro          |  PMP22_Cys109Arg        |  RHO_Cys110Tyr          |  RHO_Asp190Gly
  RHO_Pro267Thr           |  SFTPC_Leu188Gln        |  SFTPC_Leu194Pro

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 10 [PMP22_Leu16Pro, PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Gly186Cys, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Cys193Tyr, PLP1_Leu189Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Rare genetic eye disease: 10 [PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Gly186Cys, RHO_Pro267Thr, RHO_Cys110Tyr, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Cys193Tyr, RHO_Asp190Gly, PLP1_Leu189Pro]
  Genetic otorhinolaryngologic disease: 3 [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Rare chromosomal anomaly: 3 [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Ciliopathy: 3 [RHO_Pro267Thr, RHO_Cys110Tyr, RHO_Asp190Gly]
  Rare genetic respiratory disease: 2 [SFTPC_Leu194Pro, SFTPC_Leu188Gln]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: PELIZAEUS-MERZBACHER DISEASE (5x) [PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  Disease: SURFACTANT METABOLISM DYSFUNCTION, PULMONARY, 2 (2x) [SFTPC_Leu194Pro, SFTPC_Leu188Gln]
  Disease: RETINITIS PIGMENTOSA 4 (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  Disease: SPASTIC PARAPLEGIA 2, X-LINKED (2x) [PLP1_Ile152Thr, PLP1_Leu189Pro]
  inheritance_pattern: x-linked recessive (5x) [PLP1_Thr147Pro, PLP1_Asp168His, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  inheritance_pattern: autosomal dominant (4x) [PMP22_Leu16Pro, SFTPC_Leu194Pro, SFTPC_Leu188Gln, PMP22_Leu80Pro]
  inheritance_pattern: unknown (4x) [PLP1_Gly186Cys, RHO_Pro267Thr, PLP1_Cys193Tyr, PMP22_Cys109Arg]
  inheritance_pattern: autosomal recessive (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  pLDDT_Category: >90 (11x) [PMP22_Leu16Pro, PLP1_Thr147Pro, SFTPC_Leu194Pro, SFTPC_Leu188Gln, RHO_Pro267Thr, RHO_Cys110Tyr, PLP1_Thr43Ile, PLP1_Cys193Tyr, RHO_Asp190Gly, PMP22_Leu80Pro, PMP22_Cys109Arg]
  pLDDT_Category: 70-90 (4x) [PLP1_Asp168His, PLP1_Gly186Cys, PLP1_Ile152Thr, PLP1_Leu189Pro]
  RSA_Category: 10-90 (9x) [PMP22_Leu16Pro, PLP1_Thr147Pro, PLP1_Asp168His, SFTPC_Leu194Pro, SFTPC_Leu188Gln, RHO_Pro267Thr, PLP1_Thr43Ile, PMP22_Leu80Pro, PMP22_Cys109Arg]
  RSA_Category: <10 (6x) [PLP1_Gly186Cys, RHO_Cys110Tyr, PLP1_Ile152Thr, PLP1_Cys193Tyr, RHO_Asp190Gly, PLP1_Leu189Pro]
  Transmembrane (UniProt): Helical (3x) [PMP22_Leu16Pro, PMP22_Leu80Pro, PMP22_Cys109Arg]
  Topological domain (UniProt): Extracellular (3x) [RHO_Cys110Tyr, PLP1_Thr43Ile, RHO_Asp190Gly]
```







Cluster 5: Comprehensive Analysis

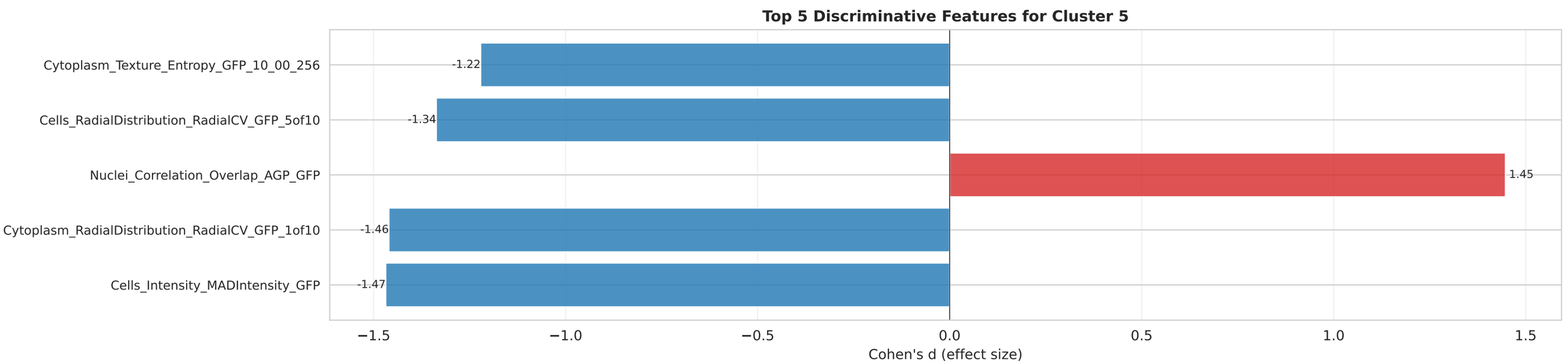
```
CLUSTER 5 ANALYSIS
================================================================================================================

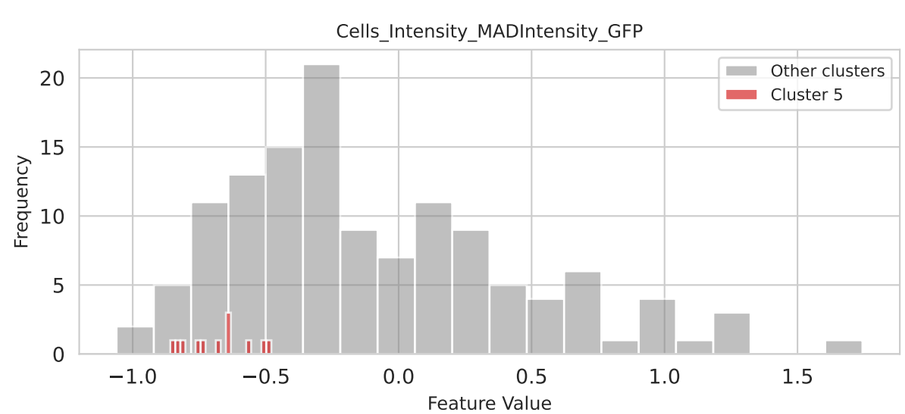
Size: 12 variants
Mean score: 0.964
Hit variants: 8 (66.7%)

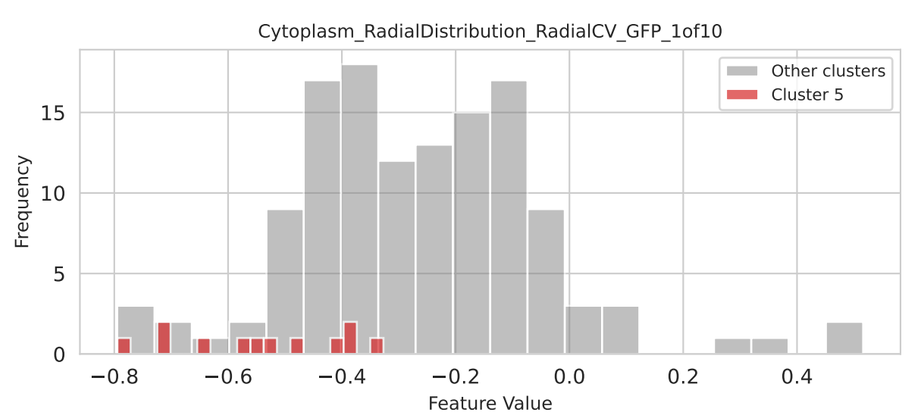
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  DCX_Arg59His            |  DCX_Arg102Cys          |  DCX_Pro191Arg          |  F9_Cys28Arg
  FBP1_Phe194Ser          |  FBP1_Gly260Arg         |  MVK_Leu255Pro          |  NT5C3A_Cys108Arg
  NT5C3A_Asp132Val        |  NT5C3A_Gly202Arg       |  NT5C3A_Gly275Arg       |  TTPA_Gly246Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare inborn errors of metabolism: 8 [FBP1_Phe194Ser, NT5C3A_Gly275Arg, MVK_Leu255Pro, FBP1_Gly260Arg, NT5C3A_Asp132Val, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg, TTPA_Gly246Arg]
  Rare genetic hematologic disease: 5 [NT5C3A_Gly275Arg, F9_Cys28Arg, NT5C3A_Asp132Val, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg]
  Rare genetic developmental defect during embryogenesis: 4 [DCX_Arg102Cys, DCX_Pro191Arg, MVK_Leu255Pro, DCX_Arg59His]
  Rare genetic neurological disorder: 4 [DCX_Arg102Cys, DCX_Pro191Arg, DCX_Arg59His, TTPA_Gly246Arg]
  Rare genetic eye disease: 2 [MVK_Leu255Pro, TTPA_Gly246Arg]
  Rare genetic systemic or rheumatologic disease: 1 [MVK_Leu255Pro]
  Rare genetic immune disease: 1 [MVK_Leu255Pro]
  Rare genetic skin disease: 1 [MVK_Leu255Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: FRUCTOSE-1,6-BISPHOSPHATASE DEFICIENCY (2x) [FBP1_Phe194Ser, FBP1_Gly260Arg]
  Disease: ANEMIA, CONGENITAL, NONSPHEROCYTIC HEMOLYTIC, 8 (2x) [NT5C3A_Gly275Arg, NT5C3A_Asp132Val]
  inheritance_pattern: autosomal recessive (5x) [FBP1_Phe194Ser, NT5C3A_Gly275Arg, FBP1_Gly260Arg, NT5C3A_Asp132Val, TTPA_Gly246Arg]
  inheritance_pattern: unknown (4x) [DCX_Arg102Cys, DCX_Pro191Arg, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg]
  pLDDT_Category: >90 (6x) [DCX_Arg102Cys, FBP1_Phe194Ser, MVK_Leu255Pro, FBP1_Gly260Arg, DCX_Arg59His, TTPA_Gly246Arg]
  pLDDT_Category: 50-70 (2x) [DCX_Pro191Arg, F9_Cys28Arg]
  RSA_Category: 10-90 (5x) [DCX_Arg102Cys, MVK_Leu255Pro, F9_Cys28Arg, DCX_Arg59His, TTPA_Gly246Arg]
  RSA_Category: <10 (2x) [FBP1_Phe194Ser, FBP1_Gly260Arg]
```







Cluster 6: Comprehensive Analysis

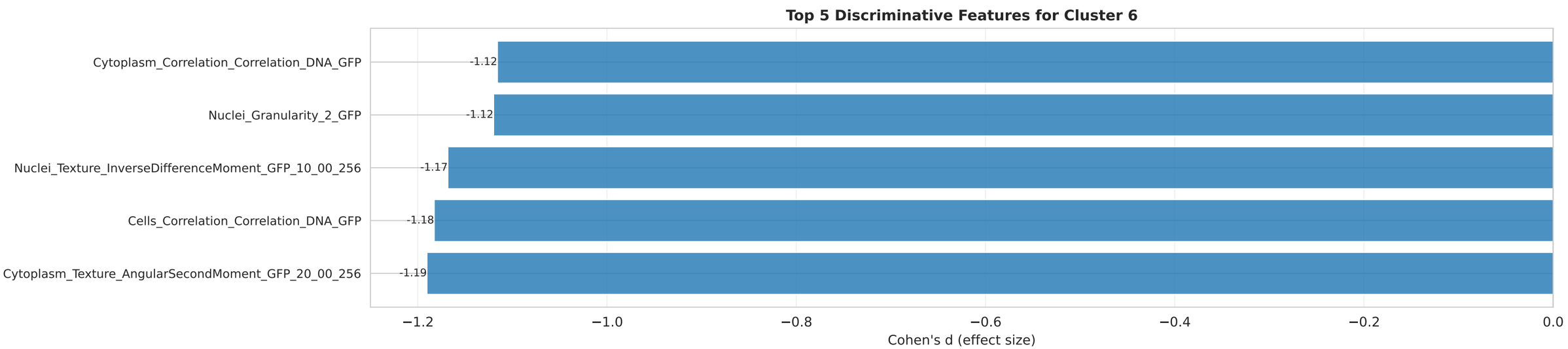
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CLUSTER 6 ANALYSIS
================================================================================================================

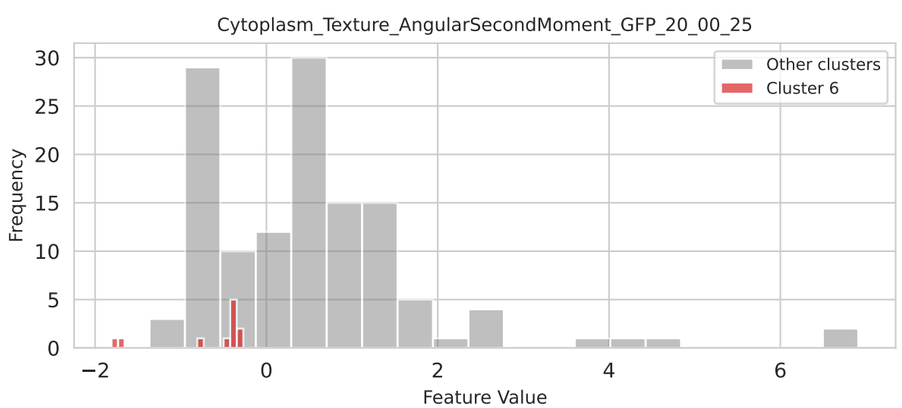
Size: 11 variants
Mean score: 0.957
Hit variants: 5 (45.5%)

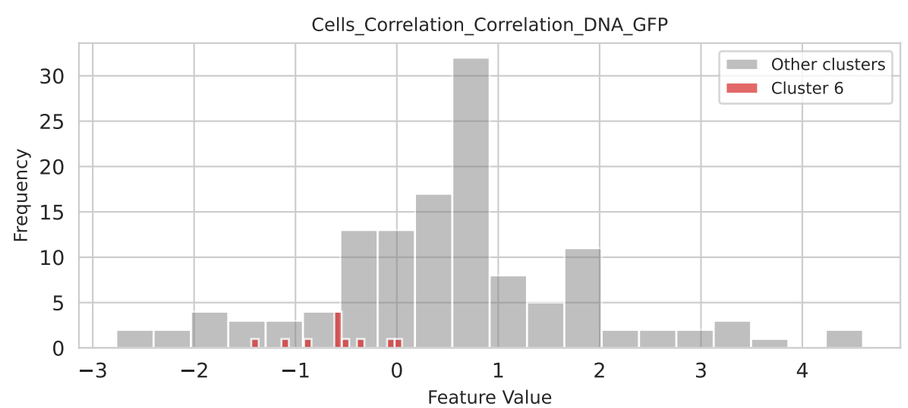
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BAG3_Leu462Pro          |  F9_Arg294Gln           |  F9_Ile316Thr           |  F9_Pro333Leu
  LITAF_Ile92Val          |  MLH1_Leu653Pro         |  NR0B1_Tyr380Asp        |  RET_Arg330Gln
  SDHD_His102Leu          |  SFTPA2_Phe198Ser       |  SFTPC_Ile73Thr

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 4 [SDHD_His102Leu, BAG3_Leu462Pro, LITAF_Ile92Val, RET_Arg330Gln]
  Rare genetic developmental defect during embryogenesis: 3 [NR0B1_Tyr380Asp, SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic endocrine disease: 3 [NR0B1_Tyr380Asp, SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic tumor: 3 [SDHD_His102Leu, MLH1_Leu653Pro, RET_Arg330Gln]
  Inherited cancer-predisposing syndrome: 3 [SDHD_His102Leu, MLH1_Leu653Pro, RET_Arg330Gln]
  Rare genetic hematologic disease: 3 [F9_Pro333Leu, F9_Arg294Gln, F9_Ile316Thr]
  Rare genetic cardiac disease: 2 [NR0B1_Tyr380Asp, BAG3_Leu462Pro]
  Rare genetic renal disease: 2 [SDHD_His102Leu, RET_Arg330Gln]
  Rare genetic respiratory disease: 1 [SFTPC_Ile73Thr]
  Rare genetic gynecological and obstetrical diseases: 1 [NR0B1_Tyr380Asp]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: HEMOPHILIA B (3x) [F9_Pro333Leu, F9_Arg294Gln, F9_Ile316Thr]
  Disease: THROMBOPHILIA, X-LINKED, DUE TO FACTOR IX DEFECT (2x) [F9_Arg294Gln, F9_Ile316Thr]
  inheritance_pattern: autosomal dominant (6x) [SFTPC_Ile73Thr, SFTPA2_Phe198Ser, SDHD_His102Leu, BAG3_Leu462Pro, LITAF_Ile92Val, RET_Arg330Gln]
  inheritance_pattern: x-linked recessive (4x) [NR0B1_Tyr380Asp, F9_Pro333Leu, F9_Arg294Gln, F9_Ile316Thr]
  pLDDT_Category: >90 (9x) [SFTPA2_Phe198Ser, SDHD_His102Leu, F9_Pro333Leu, F9_Arg294Gln, MLH1_Leu653Pro, BAG3_Leu462Pro, F9_Ile316Thr, LITAF_Ile92Val, RET_Arg330Gln]
  RSA_Category: 10-90 (6x) [SFTPC_Ile73Thr, SDHD_His102Leu, F9_Arg294Gln, BAG3_Leu462Pro, LITAF_Ile92Val, RET_Arg330Gln]
  RSA_Category: <10 (5x) [NR0B1_Tyr380Asp, SFTPA2_Phe198Ser, F9_Pro333Leu, MLH1_Leu653Pro, F9_Ile316Thr]
```







Cluster 7: Comprehensive Analysis

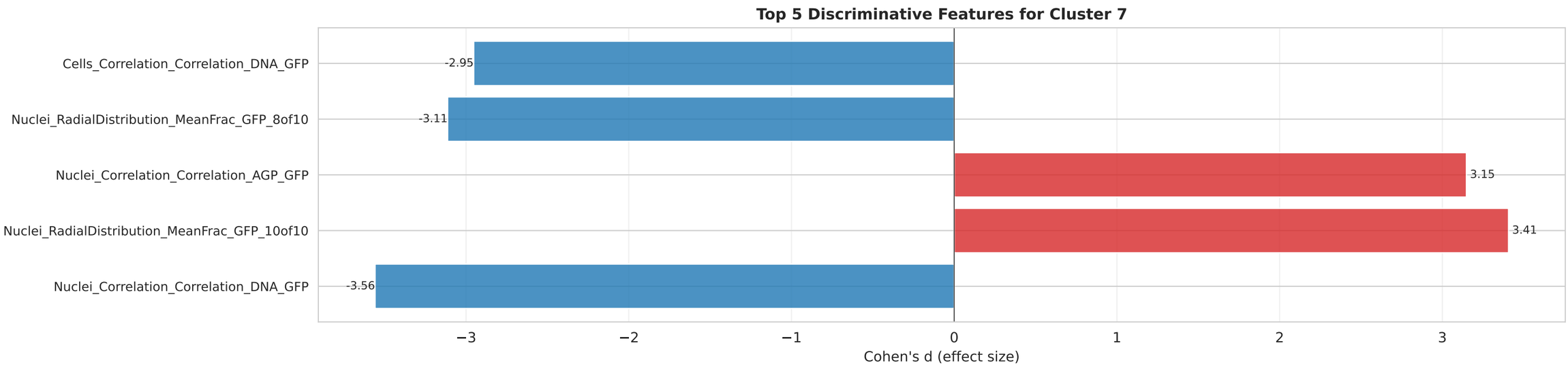
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CLUSTER 7 ANALYSIS
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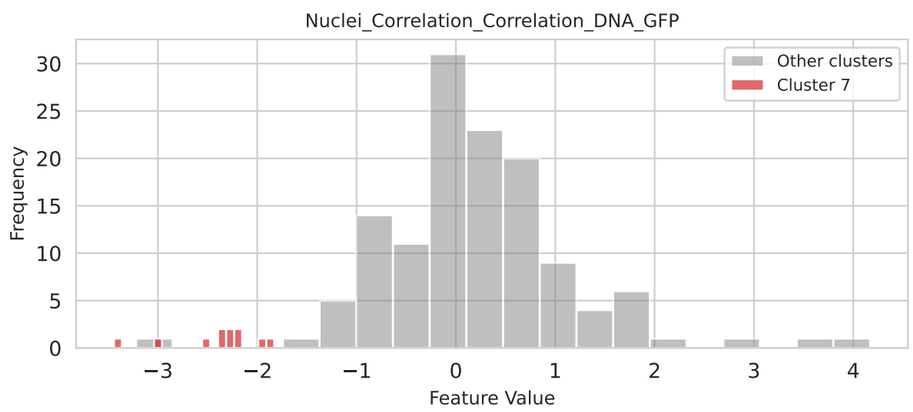
Size: 11 variants
Mean score: 0.991
Hit variants: 11 (100.0%)

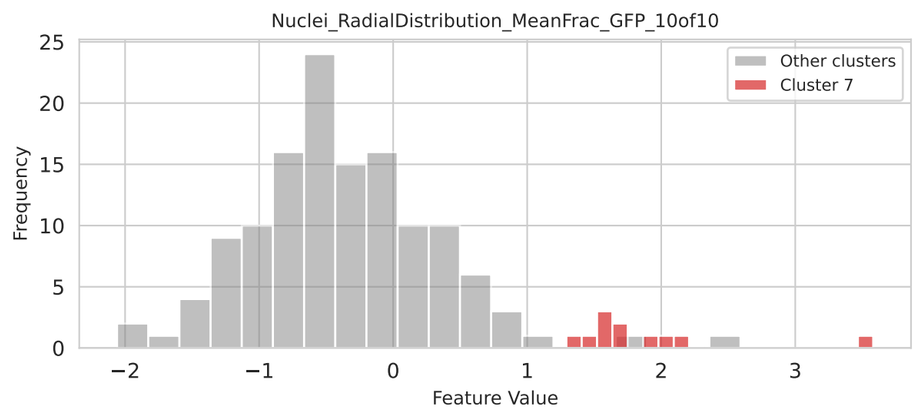
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AIPL1_Met79Thr          |  MLH1_Ile36Asn          |  MLH1_Cys39Tyr          |  MLH1_Gly147Arg
  MLH1_Val185Gly          |  MLH1_Gln701His         |  NR0B1_Arg267Pro        |  NR0B1_Arg425Ile
  NR0B1_Asn440Ile         |  SMAD3_Trp94Leu         |  SMAD3_Cys109Tyr

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic immune disease: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Rare genetic tumor: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Inherited cancer-predisposing syndrome: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Rare genetic gynecological and obstetrical diseases: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic cardiac disease: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic developmental defect during embryogenesis: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic urogenital disease: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic endocrine disease: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Genetic infertility: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic skin disease: 3 [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (4x) [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr]
  Disease: ADRENAL HYPOPLASIA, CONGENITAL (3x) [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Disease: LYNCH SYNDROME 1 (2x) [MLH1_Gln701His, MLH1_Cys39Tyr]
  inheritance_pattern: autosomal dominant (5x) [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, SMAD3_Trp94Leu, MLH1_Cys39Tyr]
  inheritance_pattern: x-linked recessive (3x) [NR0B1_Arg425Ile, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  inheritance_pattern: unknown (3x) [SMAD3_Cys109Tyr, MLH1_Ile36Asn, AIPL1_Met79Thr]
  pLDDT_Category: >90 (9x) [NR0B1_Arg425Ile, MLH1_Val185Gly, NR0B1_Asn440Ile, MLH1_Gly147Arg, SMAD3_Trp94Leu, SMAD3_Cys109Tyr, MLH1_Ile36Asn, MLH1_Cys39Tyr, AIPL1_Met79Thr]
  RSA_Category: <10 (7x) [NR0B1_Arg425Ile, MLH1_Val185Gly, MLH1_Gly147Arg, SMAD3_Cys109Tyr, MLH1_Ile36Asn, MLH1_Cys39Tyr, AIPL1_Met79Thr]
  RSA_Category: 10-90 (3x) [NR0B1_Asn440Ile, NR0B1_Arg267Pro, SMAD3_Trp94Leu]
```







Cluster 8: Comprehensive Analysis

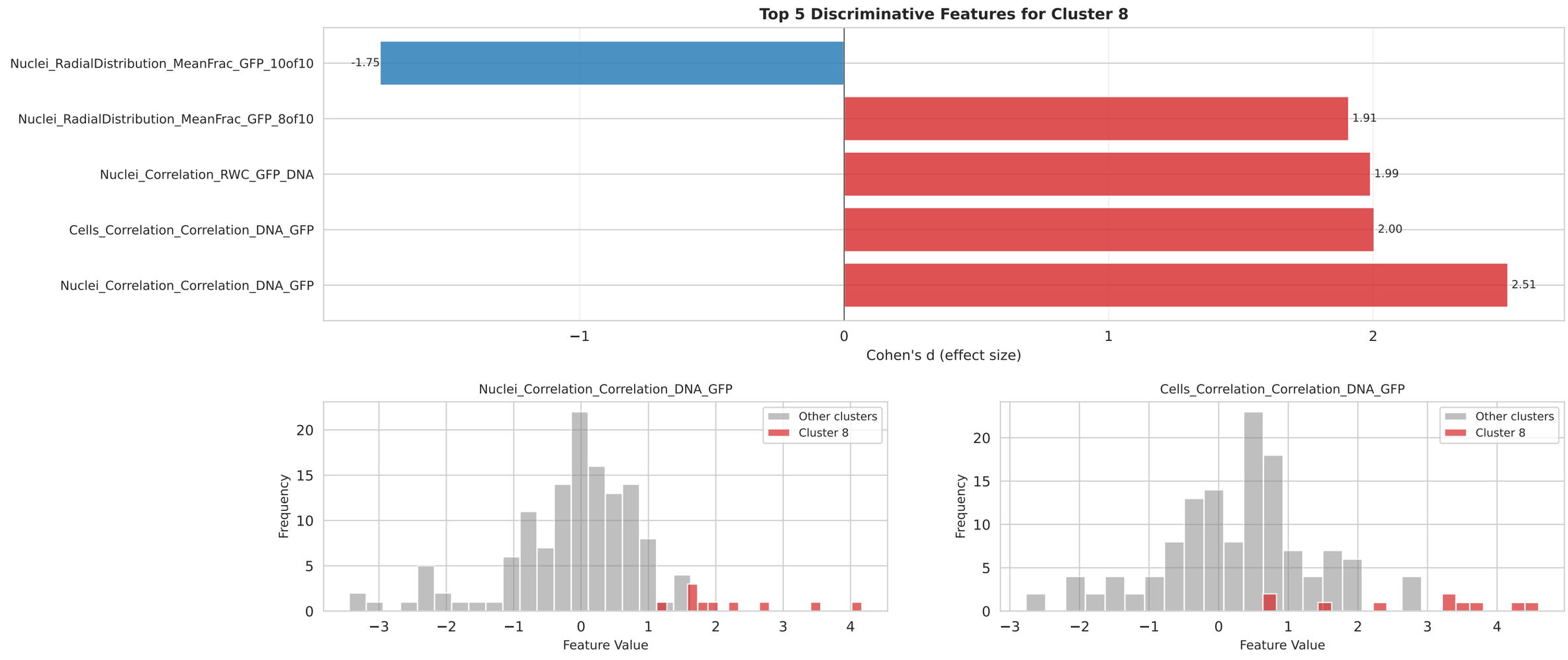
```
CLUSTER 8 ANALYSIS
================================================================================================================

Size: 10 variants
Mean score: 0.978
Hit variants: 10 (100.0%)

ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BRCA1_Glu455Lys         |  FXN_Met1Ile             |  HPRT1_Phe74Leu          |  IL36RN_Leu27Pro
  PLP1_Trp128Arg          |  PNPO_Arg95Cys           |  PNPO_Arg95His           |  PNPO_Arg116Gln
  SMARCB1_Arg37His        |  SMARCB1_Pro48Leu

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 8 [FXN_Met1Ile, PNPO_Arg95Cys, PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu, HPRT1_Phe74Leu]
  Rare genetic skin disease: 4 [IL36RN_Leu27Pro, BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare inborn errors of metabolism: 4 [PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln, HPRT1_Phe74Leu]
  Rare genetic developmental defect during embryogenesis: 3 [BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic bone disease: 3 [BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic tumor: 3 [BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic eye disease: 2 [FXN_Met1Ile, PLP1_Trp128Arg]
  Rare genetic renal disease: 2 [BRCA1_Glu455Lys, HPRT1_Phe74Leu]
  Rare genetic hematologic disease: 2 [BRCA1_Glu455Lys, HPRT1_Phe74Leu]
  Rare genetic cardiac disease: 1 [FXN_Met1Ile]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: PYRIDOXAMINE 5-PRIME-PHOSPHATE OXIDASE DEFICIENCY (3x) [PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln]
  inheritance_pattern: autosomal recessive (5x) [FXN_Met1Ile, IL36RN_Leu27Pro, PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln]
  inheritance_pattern: autosomal dominant (2x) [SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  pLDDT_Category: >90 (7x) [IL36RN_Leu27Pro, PNPO_Arg95Cys, PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  pLDDT_Category: ≤50 (2x) [FXN_Met1Ile, BRCA1_Glu455Lys]
  RSA_Category: 10-90 (6x) [BRCA1_Glu455Lys, PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  RSA_Category: <10 (2x) [IL36RN_Leu27Pro, PLP1_Trp128Arg]
```



Cluster 9: Comprehensive Analysis

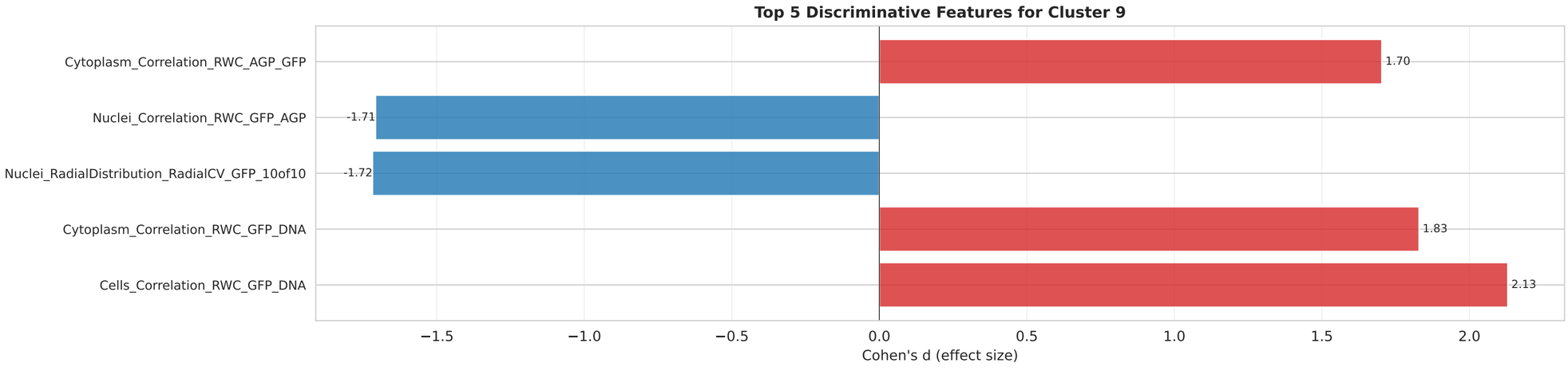
```
CLUSTER 9 ANALYSIS
================================================================================================================

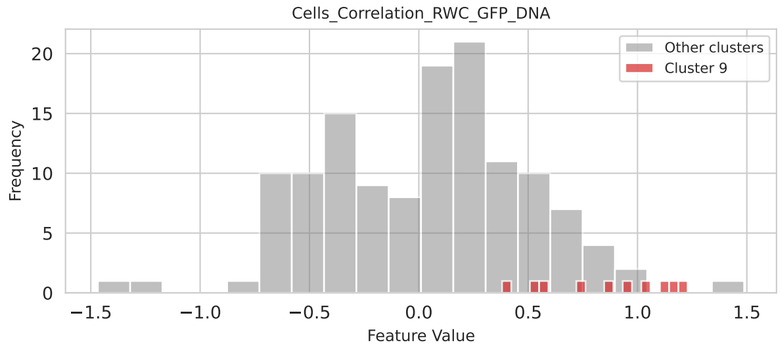
Size: 10 variants
Mean score: 0.962
Hit variants: 6 (60.0%)

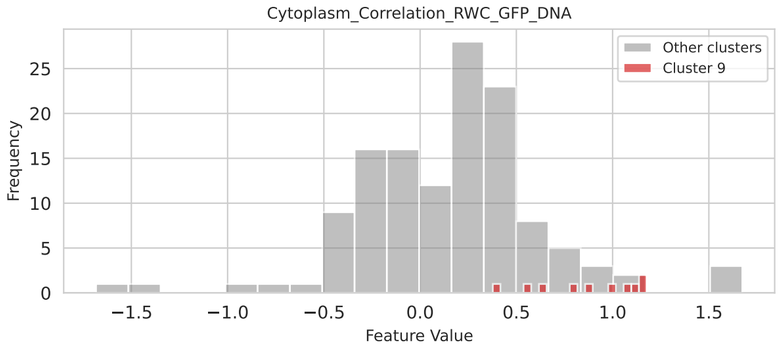
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  LITAF_Pro84Leu          |  LITAF_Trp116Gly         |  LITAF_Ser118Pro         |  LITAF_Leu122Val
  LITAF_Pro135Thr         |  LITAF_Pro135Ser         |  LITAF_Pro135Arg         |  LITAF_Phe136Leu
  LITAF_Val144Met         |  PLP1_Gly74Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 10 [LITAF_Trp116Gly, LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Ser118Pro, LITAF_Val144Met, LITAF_Pro135Arg, LITAF_Phe136Leu, PLP1_Gly74Arg]
  Rare genetic eye disease: 1 [PLP1_Gly74Arg]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: CHARCOT-MARIE-TOOTH DISEASE, DEMYELINATING, TYPE 1C (7x) [LITAF_Trp116Gly, LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Val144Met, LITAF_Pro135Arg]
  inheritance_pattern: autosomal dominant (7x) [LITAF_Trp116Gly, LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Val144Met, LITAF_Pro135Arg]
  inheritance_pattern: unknown (3x) [LITAF_Ser118Pro, LITAF_Phe136Leu, PLP1_Gly74Arg]
  pLDDT_Category: 70-90 (8x) [LITAF_Trp116Gly, LITAF_Pro135Thr, LITAF_Leu122Val, LITAF_Pro135Ser, LITAF_Pro84Leu, LITAF_Ser118Pro, LITAF_Pro135Arg, LITAF_Phe136Leu]
  RSA_Category: 10-90 (5x) [LITAF_Trp116Gly, LITAF_Leu122Val, LITAF_Pro84Leu, LITAF_Ser118Pro, LITAF_Phe136Leu]
  RSA_Category: <10 (4x) [LITAF_Pro135Thr, LITAF_Pro135Ser, LITAF_Val144Met, LITAF_Pro135Arg]
```







Cluster 10: Comprehensive Analysis

```
CLUSTER 10 ANALYSIS
================================================================================================================

Size: 9 variants
Mean score: 0.983
Hit variants: 1 (11.1%)

ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  DCX_Ser129Leu           |  DES_Asn342Asp           |  DES_Arg355Pro           |  MVK_Leu41Pro
  MVK_Ser329Asn           |  RP2_Cys86Tyr            |  RP2_Leu253Arg           |  TAT_Arg119Trp
  TAT_Leu273Pro

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic eye disease: 6 [MVK_Leu41Pro, TAT_Arg119Trp, MVK_Ser329Asn, RP2_Leu253Arg, RP2_Cys86Tyr, TAT_Leu273Pro]
  Rare inborn errors of metabolism: 4 [MVK_Leu41Pro, TAT_Arg119Trp, MVK_Ser329Asn, TAT_Leu273Pro]
  Rare genetic skin disease: 4 [MVK_Leu41Pro, TAT_Arg119Trp, MVK_Ser329Asn, TAT_Leu273Pro]
  Rare genetic developmental defect during embryogenesis: 3 [MVK_Leu41Pro, DCX_Ser129Leu, MVK_Ser329Asn]
  Rare genetic neurological disorder: 3 [DCX_Ser129Leu, DES_Asn342Asp, DES_Arg355Pro]
  Rare genetic systemic or rheumatologic disease: 2 [MVK_Leu41Pro, MVK_Ser329Asn]
  Rare genetic immune disease: 2 [MVK_Leu41Pro, MVK_Ser329Asn]
  Rare genetic cardiac disease: 2 [DES_Asn342Asp, DES_Arg355Pro]
  Ciliopathy: 2 [RP2_Leu253Arg, RP2_Cys86Tyr]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: POROKERATOSIS 3, MULTIPLE TYPES (2x) [MVK_Leu41Pro, MVK_Ser329Asn]
  Disease: MYOPATHY, MYOFIBRILLAR, 1 (2x) [DES_Asn342Asp, DES_Arg355Pro]
  inheritance_pattern: unknown (4x) [DCX_Ser129Leu, RP2_Leu253Arg, RP2_Cys86Tyr, TAT_Leu273Pro]
  inheritance_pattern: multiple (3x) [DES_Asn342Asp, DES_Arg355Pro, MVK_Ser329Asn]
  pLDDT_Category: >90 (6x) [MVK_Leu41Pro, DES_Arg355Pro, TAT_Arg119Trp, MVK_Ser329Asn, RP2_Leu253Arg, RP2_Cys86Tyr]
  pLDDT_Category: 70-90 (2x) [DCX_Ser129Leu, DES_Asn342Asp]
  RSA_Category: <10 (5x) [MVK_Leu41Pro, DCX_Ser129Leu, MVK_Ser329Asn, RP2_Leu253Arg, RP2_Cys86Tyr]
  RSA_Category: 10-90 (3x) [DES_Asn342Asp, DES_Arg355Pro, TAT_Arg119Trp]
```

